

Welcome to Machine Learning for Business Analytics IS 670

Th 7:00PM – 9:45 PM

Mostafa Amini



Course Description

This course introduces

- Basic concepts, fundamental information theories and methodologies in machine learning and business analytics
- Core techniques/algorithms for data analytics, including classification, numeric prediction, clustering, association rule mining and advanced data mining topics
- Hands on practices of major data mining techniques with real-world business applications

Mode of Delivery

- This course is conducted through **Face-to-Face** Instruction. We will meet every **Thursday 7:00 PM – 9:45 PM, COB Room 139.**

Materials

- Data Mining for Business Analytics: Concepts, Techniques and Applications in Python, by Galit Shmueli etc. (ISBN: 1119549841)
- Textbook (Optional): *Machine Learning with R*, Third Edition, by Brett Lantz, 2019. ISBN: 1788295862.
- Textbook (Optional): Business Intelligence, Analytics, + Data Science, Fourth Edition, by Sharda, Delen, and Turban, 2017
- Course materials are available on Canvas
- Google Colab for Python programming

Technical Assistance

- We will use Canvas for this course.
- If you need technical assistance at any time during the course or need to report a problem with Canvas, please contact the Technology Help Desk using their online form - [AT Help Form Homepage](#) or by phone at (562) 985-4959 or visit them on campus in the Horn Center or in the Library.

About This Course

- This is basically an introduction to machine learning, business analytics, and data mining, where you will be exposed to a number of different concepts and learn to combine them to solve business problems
 - This is NOT a computer science class, but you will be required to write code and likely learn new syntax
 - We will NOT take an in-depth dive into any one principle or method

Grading

	Points	Percentage
2 Exams (75 points each)	150	30%
2 Quizzes (25 points each)	50	10%
11 Lab Assignments (10 points each, drop the lowest)	100	20%
4 Homework Assignments (25 points each)	100	20%
Group Project	100	20%
Total	500	100%

***Notes:**

1. The number of assignments for each category may change.
2. Detailed description of each assignment will be available on Canvas.
3. All Quizzes and Exams are open book.

For each lab, you will get 10 points if you submit a complete lab file by the due date.

Grading Scale

Percent Range	Letter Grade
90-100%	A
80-89%	B
70-79%	C
60-69%	D
Below 60%	F

Canvas

Dates and Schedules

- Pay attention the class dates and due dates

Announcements

- Canvas announcements will be made from time to time, please pay attention to them

The “week” module

- It is very important to check the "week" module and be clear about the weekly tasks and due days

Communication

Primary communication method will be email

- Mostafa.amini@csulb.edu

Office Hours

- Tu 4:15PM - 5:15PM
- **Office:** COB-440

Attendance

- No portion of your grade will be tied directly to attendance, however, class meetings and participation are vital components of this course
- A large portion of your grade is based on in classroom activities
 - Quizzes and Exams will be completed on class
 - All material covered in class is expected to be used on the exams
 - Labs

Assignments

- All homework and lab assignments are due at 11:59 pm on the due date as posted in Canvas
- For late assignments (up to 3 days late), the final assignment score will be dropped by 30%
- **Do not wait** until the last moment to submit on Canvas

Quizzes and Exams

- All Quizzes and Exams are open book
- Quizzes and Exams include
 - Part1: True/False and multiple choice questions
 - Part2: Short answers
 - Part3: Code writing – you will be required to write Python code on Canvas.

AI Policy

- Artificial Intelligence (AI) can play a significant role in enhancing learning and facilitating complex tasks.
- AI tools can provide valuable assistance in understanding difficult concepts, debugging code, and generating ideas.
- It is crucial to use these tools responsibly and ethically to maintain academic integrity and ensure that learning objectives are met. Therefore, AI use is permissible under certain conditions.

AI Policy

Appropriate Use:

- **Learning and Assistance:** Students are encouraged to use AI tools, such as ChatGPT, for learning purposes, to get clarifications on complex topics, and to assist with coding tasks. For example, using AI to debug code or to get explanations of machine learning concepts is appropriate.
- **Research and Development:** AI tools can be used for generating ideas, exploring datasets, and developing models, provided that the final work reflects the student's understanding and effort.
- **Collaboration:** AI can be used as a collaborative tool to brainstorm project ideas or to get suggestions for improving existing work.

Inappropriate Use:

- **Plagiarism:** Submitting AI-generated content as one's own without proper citation is strictly prohibited. This includes copying code, text, or research findings directly from AI tools.
- **Exam and Assignment Integrity:** Using AI to complete assignments and projects in a way that undermines academic integrity is not allowed. For instance, generating a chunk of project reports or writing code for assignments using AI without personal input is considered cheating.

Inappropriate Use of AI

- First Offense: The student will receive a warning and be required to redo the assignment or project using their own understanding and effort.
- Second Offense: The student will receive a zero for the assignment or project in question.
- Repeated Offenses: Continued inappropriate use of AI tools will result in further disciplinary action, which may include failing the course and/or reporting to the academic integrity board.

Question?

Lecture 1: Introduction to Business Analytics

Overview

- What is Business Data Analytics
- Why Business Analytics
- Business Data Analytics Process
- Business Analytics Applications
- Business Analytics Trends

What is Business Data Analytics

- Business Data Analytics (BA or BDA) focuses on extracting meaningful insights/patterns from (large) datasets (data mining) using statistics, machine learning, and analytical tools to support data-driven business decisions.
- BDA includes data collection, data aggregation, analysis, and meaningful presentation that facilitates decision-making

What is Business Data Analytics

- Data knows more about your music preference



What is Business Data Analytics

- Image classification

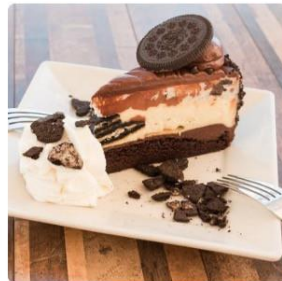
Photos for The Cheesecake Factory



 Add photos

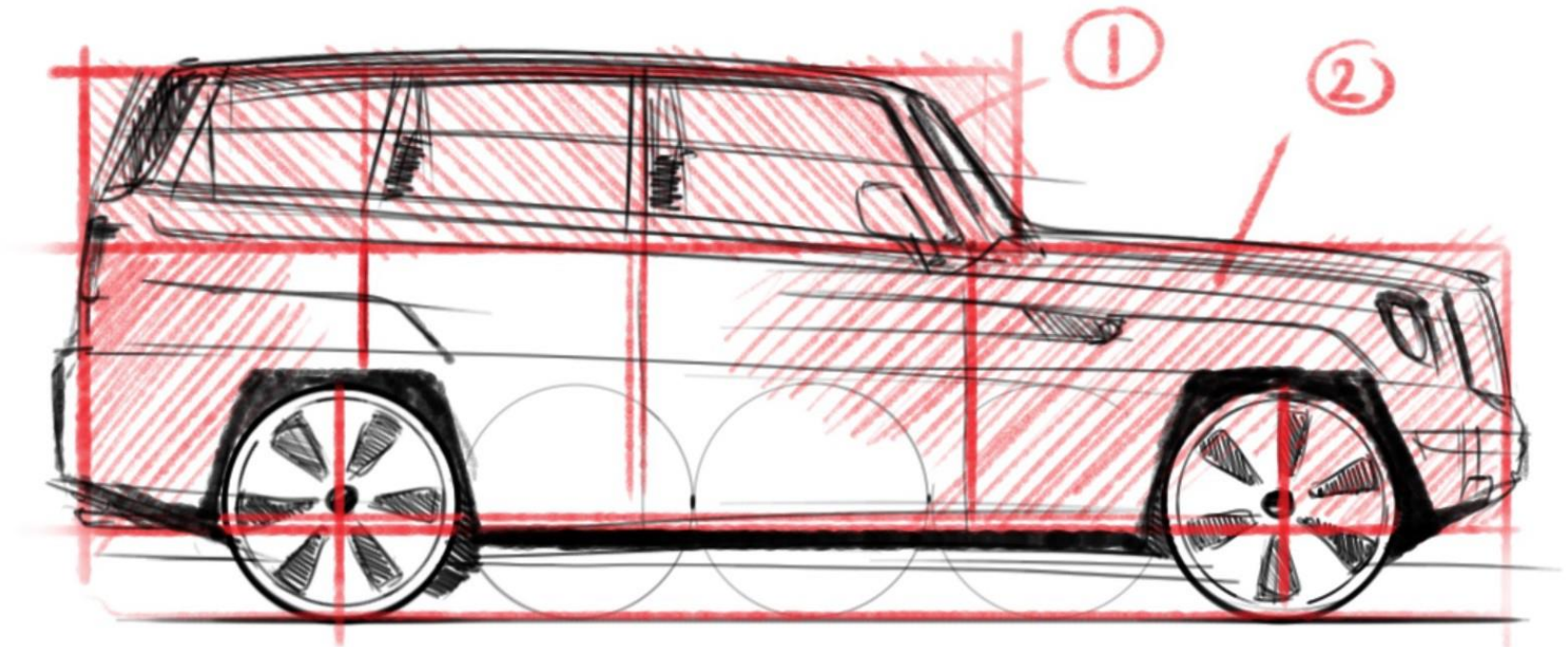
< **All (537)** Food (453) Inside (29) Drink (20) Menu (8) Outside (7)

> Search photos...



What is Business Data Analytics

- Car manufacturer improved vehicle models via the social media platform



What is Business Data Analytics

- Tesla grants certain customers early access to a “feature complete” version of the company’s “full self-driving” (FSD) capabilities in 2020.
 - The car is able to drive from someone’s home to their work without intervention.

How self-driving cars work?

- Self-driving cars combine a variety of sensors to perceive their surroundings.
- Advanced control systems interpret sensory information to identify appropriate navigation paths, as well as obstacles and relevant signage.



Why Business Data Analytics?

- Data: any facts, numbers, or text that can be processed by a computer.
 - Computerization of businesses produce huge amount of data
 - Online e-businesses are generating even larger data sets

Why Business Data Analytics?

- Data



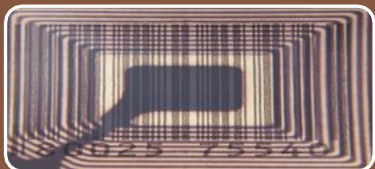
Business data – e.g., demographic data, transactional databases, and customer behavior data



Web data – e.g., page content, page linkage and clickstream



System data – e.g., service logs, quality and availability of network, computer, and manufacturing devices



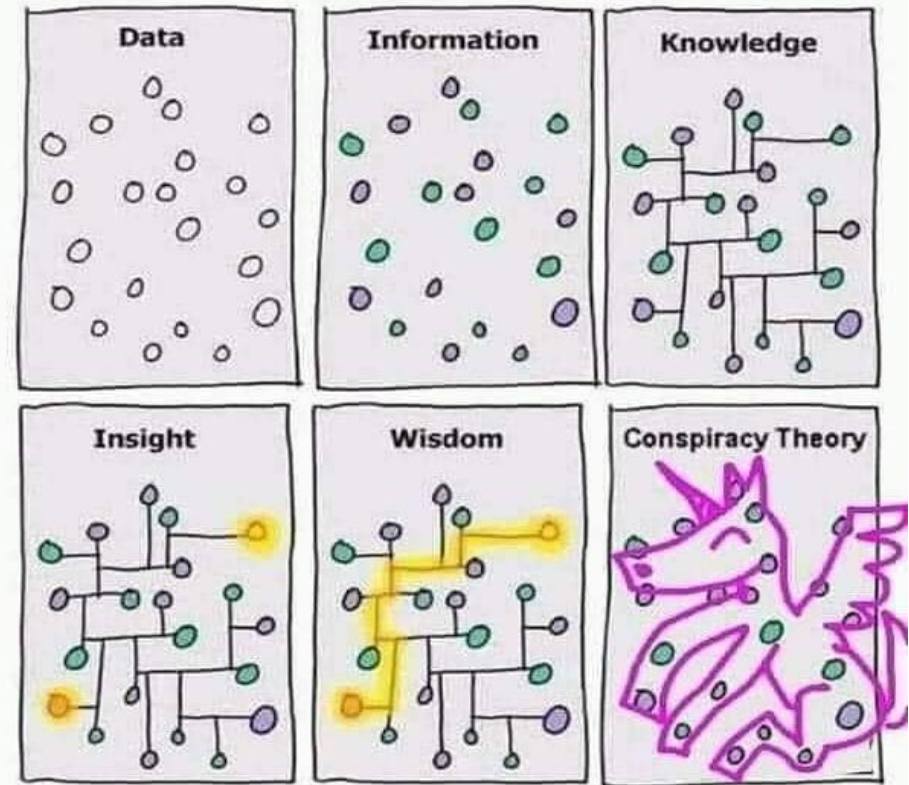
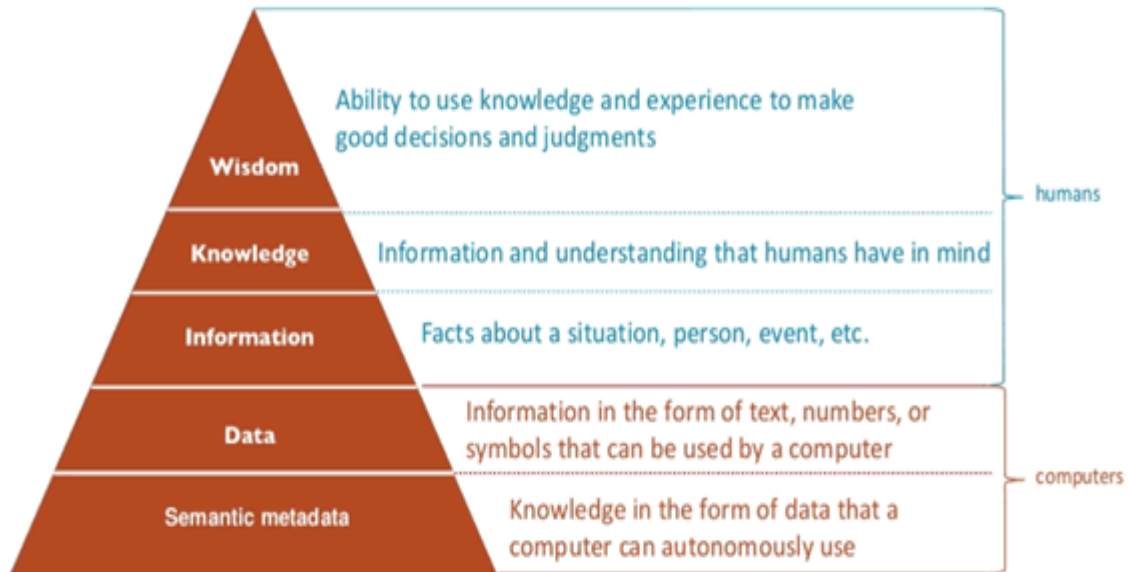
Sensor data – e.g. RFID devices for homeland security, supply-chain and environmental applications

Why Business Data Analytics?

- There is a huge gap from stored data to knowledge

DIKW paradigm today

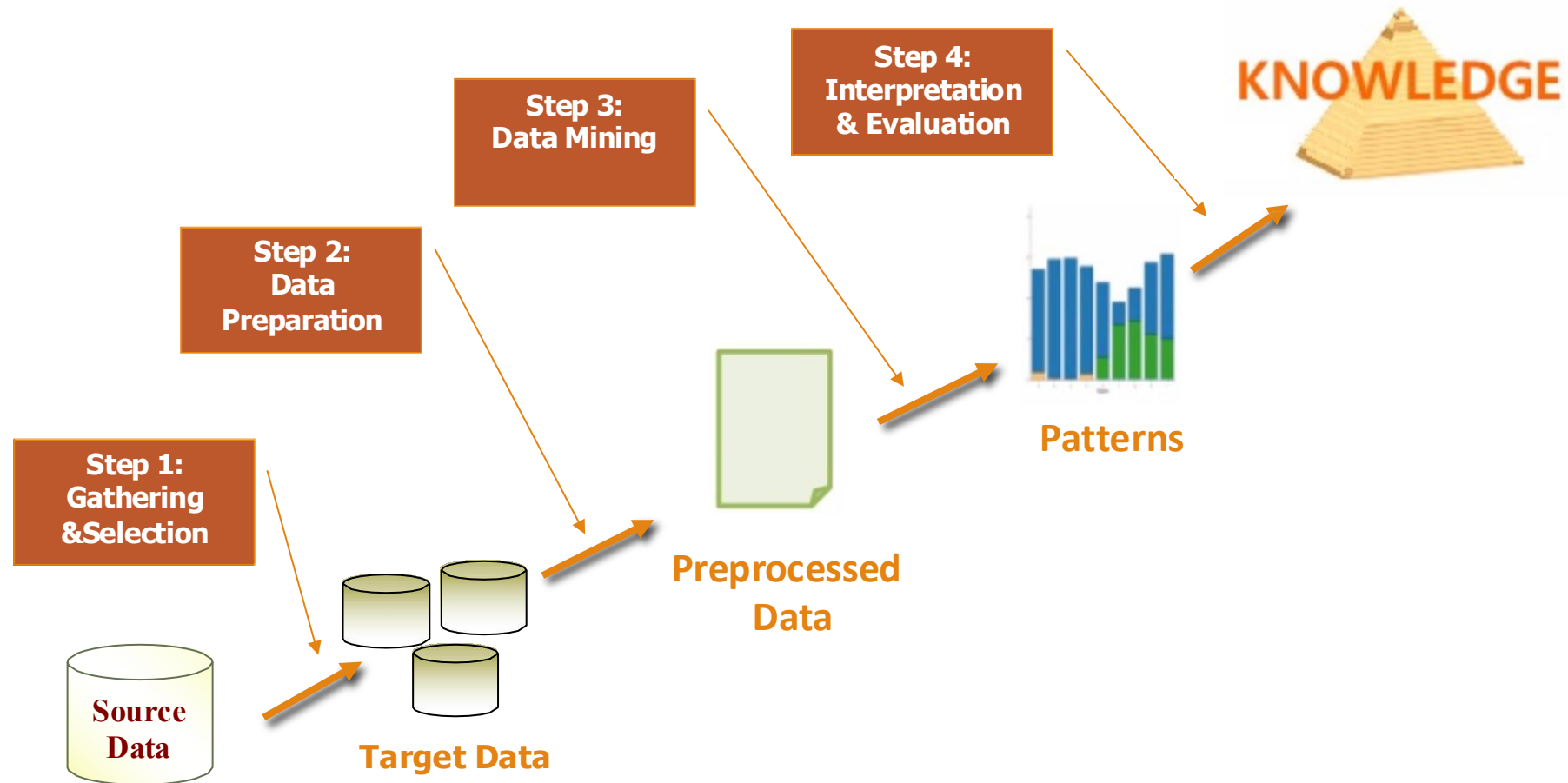
Adding understandable data to the stack



Why Business Data Analytics?

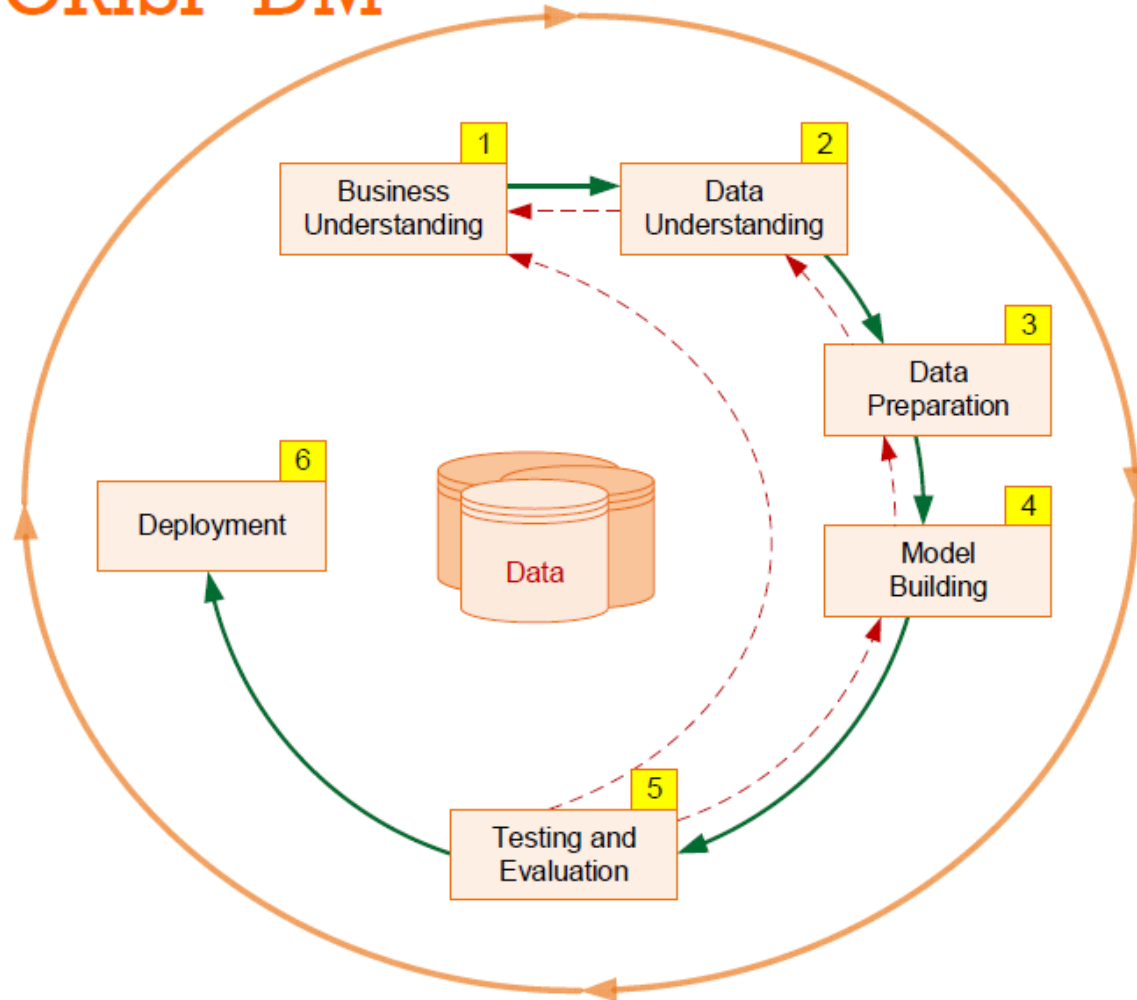
- Business intelligence helps generate insights from data to support business decisions and improve the profitability and efficiency

Business Data Analytics Process



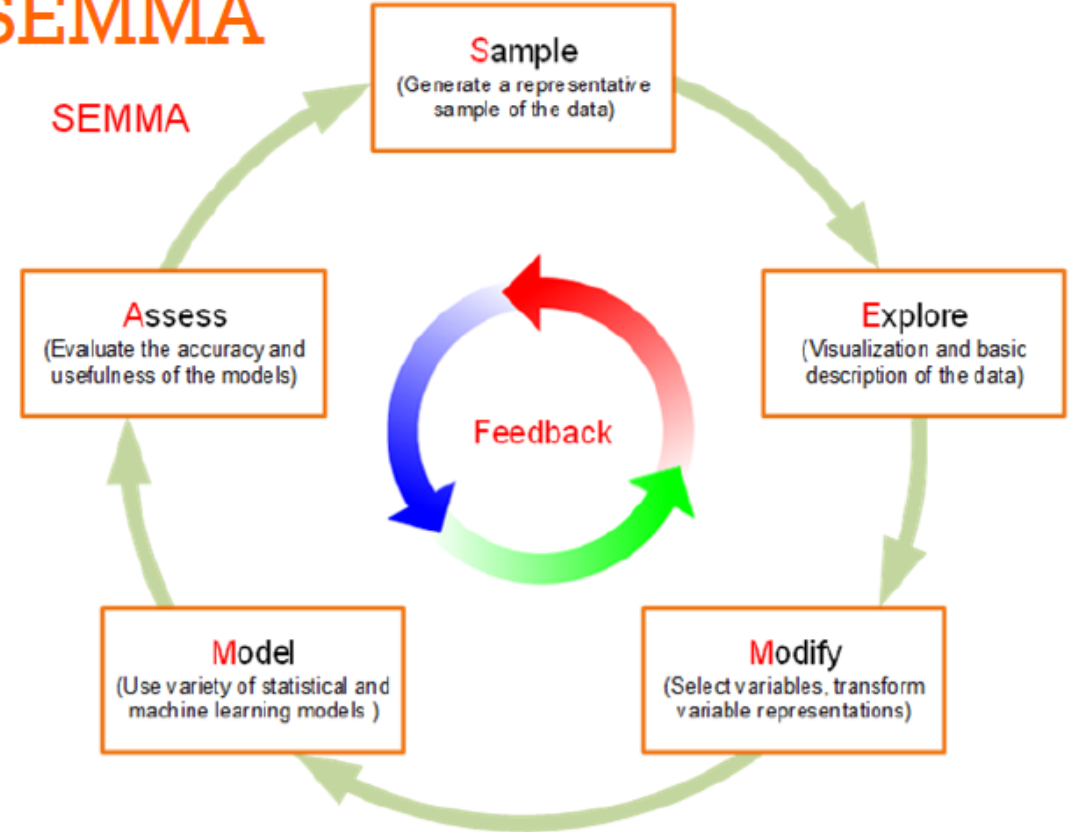
BUSINESS ANALYTICS PROCESS

CRISP-DM



SEMMA

SEMMA



Business Data Analytics Process

- **Data understanding:** With preliminary analysis, data exploration provides a high-level overview of each variable in the data set and interaction between the variables.

features

The diagram shows a table with 6 columns and 10 rows. A bracket above the columns is labeled 'features'. A bracket to the right of the rows is labeled 'examples'.

year	model	price	mileage	color	transmission
2011	SEL	21992	7413	Yellow	AUTO
2011	SEL	20995	10926	Gray	AUTO
2011	SEL	19995	7351	Silver	AUTO
2011	SEL	17809	11613	Gray	AUTO
2012	SE	17500	8367	White	MANUAL
2010	SEL	17495	25125	Silver	AUTO
2011	SEL	17000	27393	Blue	AUTO
2010	SEL	16995	21026	Silver	AUTO
2011	SES	16995	32655	Silver	AUTO

If a variable represents a characteristic measured in numbers, it is unsurprisingly called **numeric variable**.

if a variable consists of a set of categories, the variable is called **categorical variable**.

Business Data Analytics Process

- **Data preparation:** Before applying the data mining algorithm, we need to prepare the data set for handling of any anomalies that may be present in the data.
 - Finding outliers; e.g., patients with age >120
 - Missing values; data sparsity
 - Duplicate
 - Highly correlated variables; age/DOB
 - Data cleaning and transforming

Business Data Analytics Process

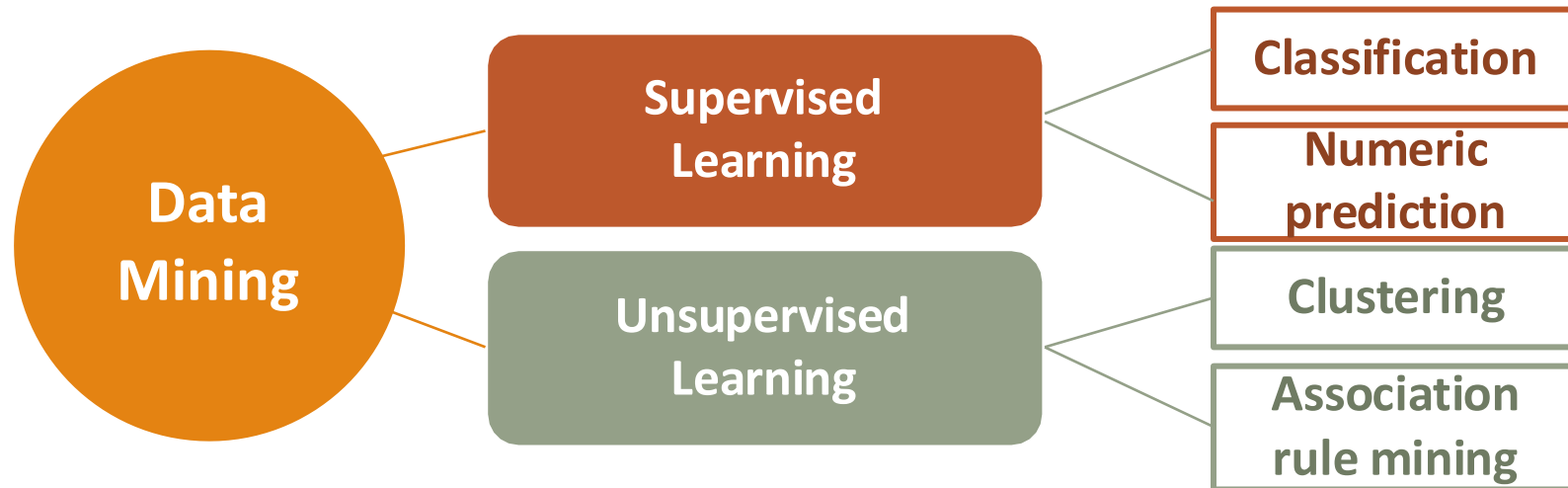
- **Data mining tasks**
 - Classification
 - Numeric prediction
 - Clustering
 - Association rule mining

Business Data Analytics Process

- **Interpretation and evaluation:**

- Because each data mining model results in a biased solution to the learning problem, it is important to evaluate how well the algorithm learns from its experience.
- Depending on the type of model used, you might be able to evaluate the accuracy of the model using a test dataset or you may need to develop measures of performance specific to the intended application.

Data Mining Tasks



Data Mining Tasks

- **Supervised learning** is where you have input variables (x) and an output variable (Y) and you use an algorithm to learn the mapping function from the input to the output.
 - $Y=f(X)$; $X \xrightarrow{\text{predict}} Y$



The supervision does not refer to human involvement, but rather to the fact that the target values provide a way for the learner to know how well it has learned the desired task.



A **predictive model** is used for tasks that involve, as the name implies, the prediction of one value using other values in the dataset.

age	sex	bmi	children	smoker	region	expenses	InsurancePlan
18	male	23.2	0	no	southeast	1121.87	Basic
18	male	30.1	0	no	southeast	1131.51	Basic
18	male	33.3	0	no	southeast	1135.94	Basic
18	male	33.7	0	no	southeast	1136.4	Basic

Data Mining Tasks

- **Supervised learning-Classification**

- A classification problem is to predict which category an example belongs to
 - Output/Target variable is a category
 - An e-mail message is spam
 - A person has cancer
 - A football team will win or lose
 - An applicant will default on a loan

Data Mining Tasks

- **Supervised learning-Classification**

- Predicting Pokémon battle winner with classification



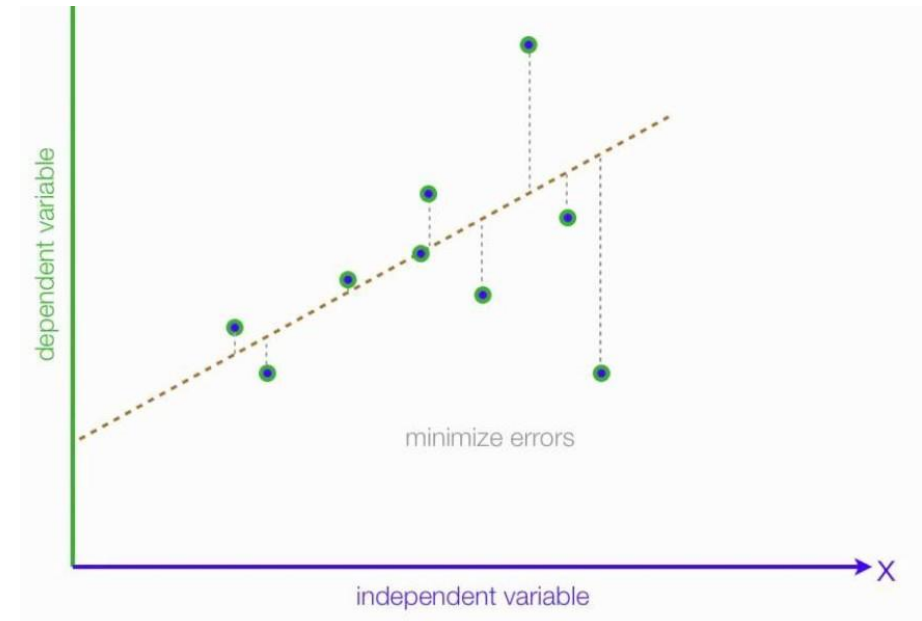
Image Source : The Verge

Data Mining Tasks

- **Supervised learning-Numeric Prediction**

- A numeric prediction problem is when the output variable is a real value, such as “dollars” or “weight”.

- If a company’s sales have increased steadily every month for the past few years, by conducting a linear analysis on the sales data with monthly sales, the company could forecast sales in future months.



Data Mining Tasks

- **Supervised learning-Numeric Prediction**
 - Predict medical expenses

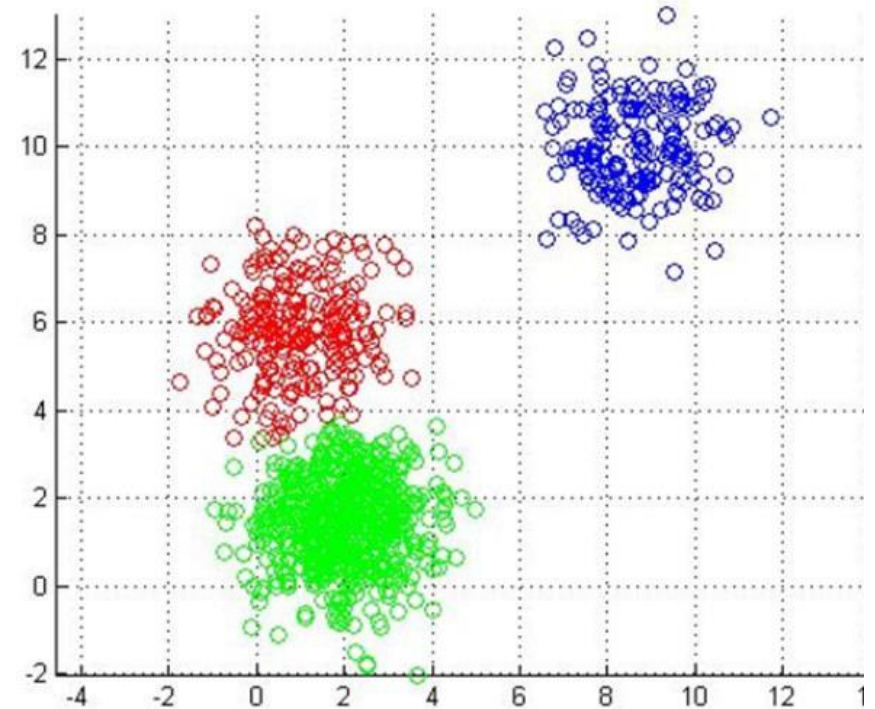
age	sex	bmi	children	smoker	region	expenses	InsurancePlan	
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18	male	33.7	0	no	southeast			

Data Mining Tasks

- **Unsupervised learning** is where you only have input data (X) and no corresponding output variables.
- A **descriptive model** is used for tasks that would benefit from the insight gained from **summarizing data in new and interesting ways**.
- There is no target to learn, the process of training a descriptive model is called unsupervised learning.

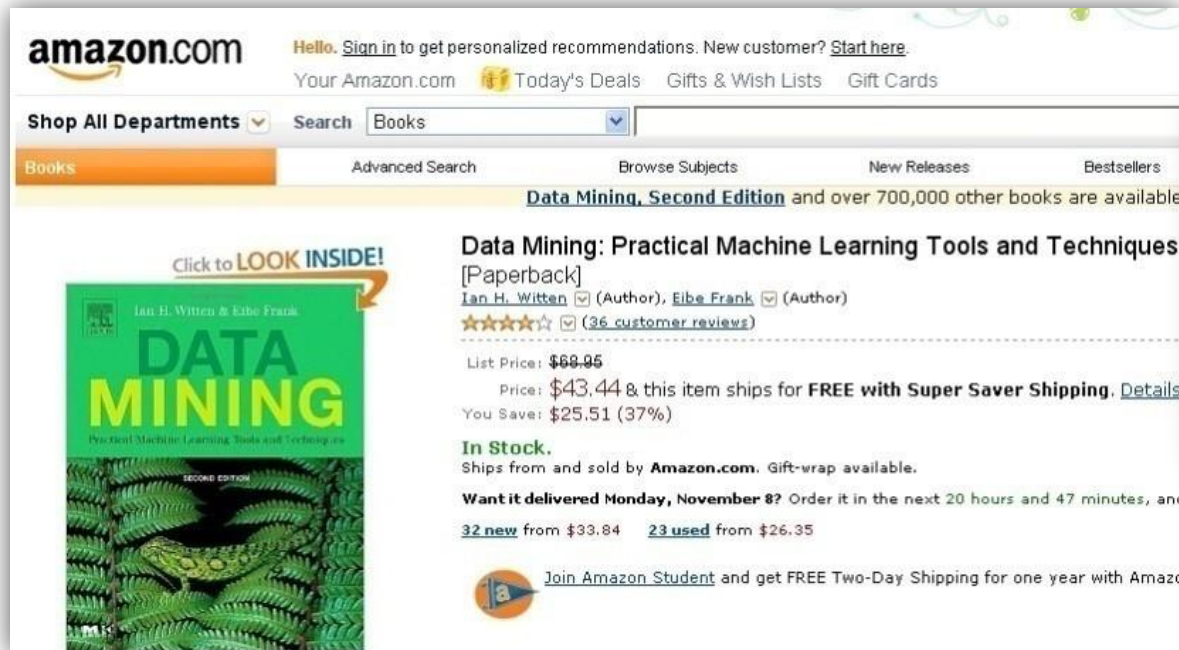
Data Mining Tasks

- **Unsupervised learning-Clustering**
 - Customer segmentation
 - Retailers want to segment customers based on their spend patterns and understand price sensitivity of the customers. Some of the segmentation variables considered are – total spend, value of discounts, number of items bought on discounts.



Data Mining Tasks

- **Unsupervised learning-Association rule mining**



Promotions to
retain customers
& increase sales

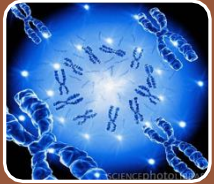
Business Analytics Applications



Predicting the rise & fall of stocks



Weather Forecasting



Gene data



Medical Diagnosis

Business Analytics Applications

- Target Marketing
- Credit Scoring
- Sales Forecasting
- Promotion Analysis
- Distribution Channel Analysis
- Customer Profiling
- Customer Profitability Analysis
- Cross Sell/Up Sell
- Help Desk Problem Resolution
- Customer Service Automation
- Network Forecasting
- Tariff Modeling
- E-government
- Fraud detection
- Security Management
- Product/Product Line Profitability
- Merchandise Planning
- Resource Management
- Operations Management
- Capacity Management
- Store/Branch Performance Analysis
- Store/Branch Site Selection
- Diagnosis decision support
- Gene and protein analysis
- Homeland Security

Business Analytics Trends

- Due to rapid growth of data, ***big data*** processing is becoming a major component of the knowledge discovering process.
 - Cloud computing technology has been emerged to manage large data sets efficiently.
 - Data analytical engines, such as Apache Spark, enable enterprises to mine their business data in the cloud.

What is Big Data?

Extremely large and complex sets of data that traditional data processing tools cannot easily store, manage, or analyze.



Volume – Massive amounts of data (e.g., terabytes or petabytes from social media, sensors, transactions).



Velocity – The speed at which new data is generated and needs to be processed (e.g., real-time stock trading).



Variety – Different types of data (structured like databases, unstructured like videos or tweets).



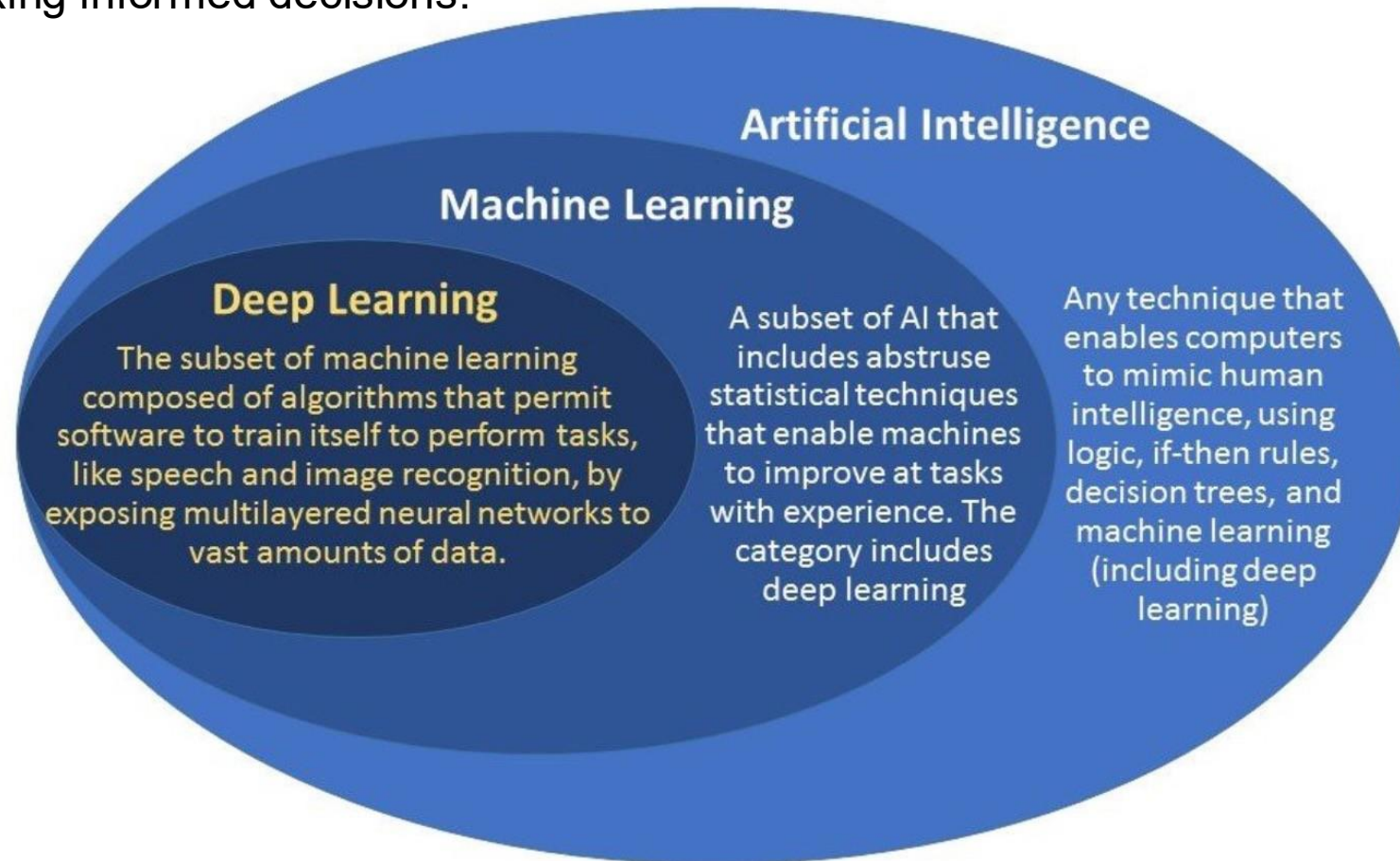
Veracity – The reliability or accuracy of the data.



Value – The potential insights or benefits that can be gained from analyzing the data.

Big Data Analytics

The process of examining **big data** to uncover **hidden patterns**, **correlations**, **trends**, and **insights** that can help in making informed decisions.



AI History

Artificial Intelligence

Machine Learning

Deep Learning

Generative AI

1943 – 1950s

Artificial Intelligence

the field of computer science that seeks to create intelligent machines that can replicate or exceed human intelligence.

1959-1997

Machine Learning

subset of AI that enables machines to learn from existing data and improve upon that data to make decisions or predictions.

2011-2017

Deep Learning

a machine learning technique in which layers of neural networks are used to process data and make decisions.

2020-Present

Generative AI

create new written, visual, and auditory content given prompts or existing data.

Business Analytics Trends

- Machines That Learn More *Effectively*
 - This will lead to developments in how algorithms are treated, such as AI deployments that can recognize, alter, and improve upon their own internal architecture with minimal human supervision.

Business Analytics Trends

- Deep Learning
 - On March 2016, Google DeepMind's deep learning program AlphaGo beat Lee Sedol, the Go World Champion
 - Recent advancements in automatic machine translation, object classification in photographs, image caption generation, and more.

Business Analytics Trends

- Deep Learning
 - Generative adversarial network (GAN) — a class of neural network — to generate some extremely realistic faces.



Business Analytics Trends

- Deep Learning
 - Generative adversarial network (GAN) — a class of neural network — to generate some extremely realistic faces.



Business Analytics Trends

- Deep Learning
 - Generative adversarial network (GAN) — a class of neural network — to generate some extremely realistic faces.
 - Large Language Models (LLM), GPT

